Area in Polar Coordinates

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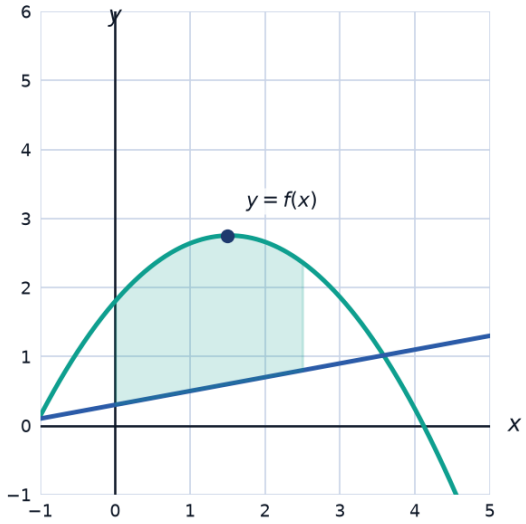
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The polar area formula**

State the formula for the area enclosed by a polar curve $r = f(\theta)$ between $\theta = \alpha$ and $\theta = \beta$.

2 **[Foundational]** • **Target: The polar area formula**

Find the area enclosed by one full loop of $r = 3\cos\theta$ (i.e. $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$).



3 **[Foundational]** • **Target: Area between two polar curves**

Find the area inside the circle $r = 3$ but outside the circle $r = 2$.

4 **[Foundational]** • **Target: Area of one petal of a rose curve**

Find the area of one petal of $r = 2\sin(2\theta)$, using the interval $\theta \in [0, \frac{\pi}{2}]$ where one petal is traced.

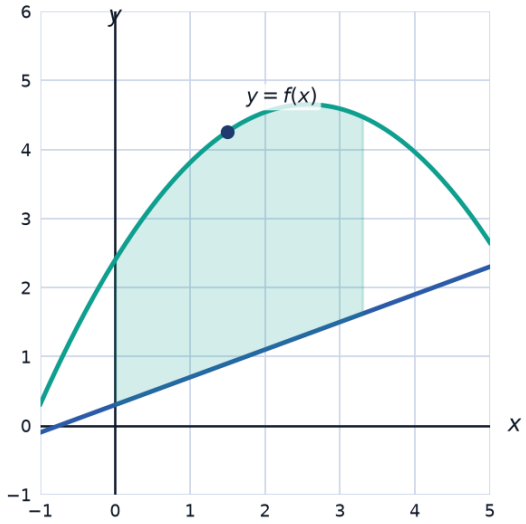
5 **[Foundational]** • **Target: Setting up area between curves that intersect**

Set up (but do not evaluate) the integral for the area inside both $r = 2$ and $r = 2 + 2\cos\theta$, noting they intersect where $2 = 2 + 2\cos\theta$, i.e. $\theta = \pm\frac{\pi}{2}$.

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6 [Foundational] • Target: Area of a full rose curve

Find the total area enclosed by all 4 petals of $r = 3\cos(2\theta)$, using the fact that one petal is traced for $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.



7 [Foundational] • Target: Area inside a limaçon

Find the area enclosed by the limaçon $r = 3 + \cos\theta$ (a curve without an inner loop, since the constant exceeds the coefficient).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Area inside a limaçon

Find the area of the small inner loop of the limaçon $r = 1 + 2\cos\theta$ (constant less than coefficient, so an inner loop exists), traced for $\theta \in \left[\frac{2\pi}{3}, \frac{4\pi}{3}\right]$.

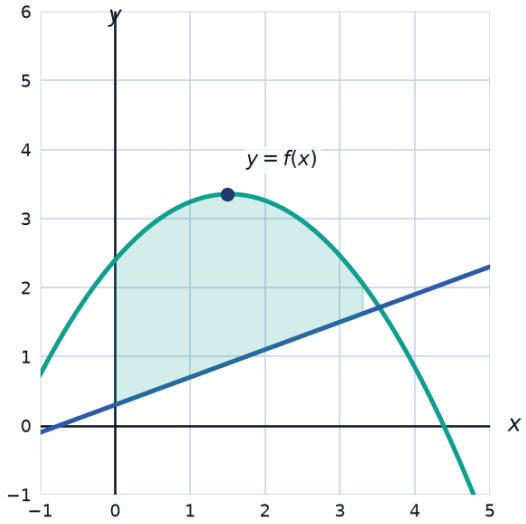
9 [Intermediate] • Target: Setting up area problems (AP-style, no evaluation)

Set up (but do not evaluate) the integral for the area inside the rose $r = 2\sin(3\theta)$ that lies in the first quadrant only, for one petal traced on $\theta \in [0, \frac{\pi}{3}]$.

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10 [Intermediate] • Target: Setting up area problems (AP-style, no evaluation)

Set up (but do not evaluate) the integral for the area between the circle $r = 5$ and the cardioid $r = 5 - 5\sin\theta$ (inside the circle, outside the cardioid), for $\theta \in [0, \pi]$.



11 [Intermediate] • Target: Finding intersection points of polar curves

Find all points of intersection of $r = 2\cos\theta$ and $r = 2\sin\theta$ (both circles of radius 1) by setting the equations equal.

12 [Intermediate] • Target: Finding intersection points of polar curves

Explain why checking intersections of two polar curves algebraically (setting $f(\theta) = g(\theta)$) can MISS the pole as an intersection point, and why it must be checked separately.

13 [Intermediate] • Target: Area using symmetry to simplify computation

Explain how the symmetry of $r = 3\cos(2\theta)$ about both axes allows the total area to be found by computing just one petal and multiplying, rather than integrating over the full range at once.

14 [Intermediate] • Target: Area review: full worked example

Find the area enclosed by $r = \sqrt{\theta}$ for $\theta \in [0, \pi]$ (a spiral arc closed off by the segment back to the pole).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

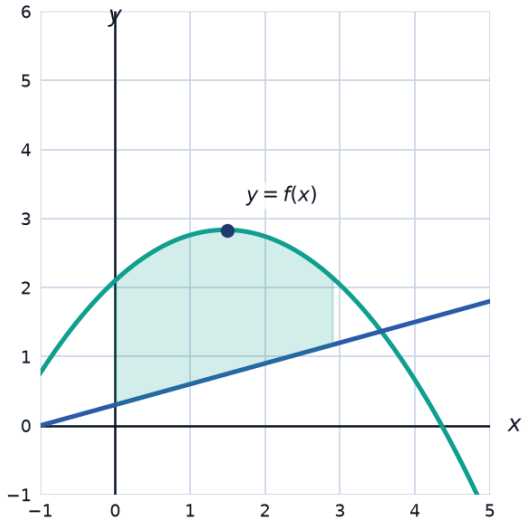
15 [Challenging] • Target: Area review: full worked example • Cross-Lesson: 1.01 Integration by Parts

Find the area enclosed by $r = 4\sin\theta$ (a circle) for $\theta \in [0, \pi]$, and verify your answer matches the known area formula for a circle of the appropriate radius.

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16 [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Explain why the polar area formula $A = \frac{1}{2}\int r^2\,d\theta$ uses r^2 rather than just r, by relating it to the area of a thin circular sector.



17 [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

A student claims the area between $r = 1$ and $r = 2$ (an annulus, full circle) is simply $\frac{1}{2}\int_0^{2\pi} (2-1)^2\,d\theta$. Explain the error and give the correct setup.

18 [Challenging] • Target: Area of a full rose curve • Cross-Lesson: 3.01 Parametric Equations

Level 1 · FoundationsConvert the Cartesian equation $x^2 + y^2 = 6y$ into polar form, and identify the resulting geometric shape.

19 [Challenging] • Target: Area inside a limaçon • Cross-Lesson: 3.02 Vector-Valued Functions

Use a two-term Maclaurin polynomial for e^x to approximate $e^{0.1}$.

20 [Challenging] • Target: Setting up area problems (AP-style, no evaluation) • Cross-Lesson: 4.01 Polar Curves and Derivatives

Level 1 · FoundationsEvaluate the improper integral $\int_1^{\infty} \frac{1}{x^3}\,dx$ using proper limit notation, or state that it diverges.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The polar area formula

Find the area enclosed by $r = 1$ for $\theta \in [0, \frac{\pi}{2}]$.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$

c. $-\frac{7\pi}{4}$

d. $\frac{3\pi}{4}$

22 [Foundational] • Target: Area of a full cardioid

Find the total area enclosed by the curve $r = 8 + 7\sin\theta$.

a. 113π

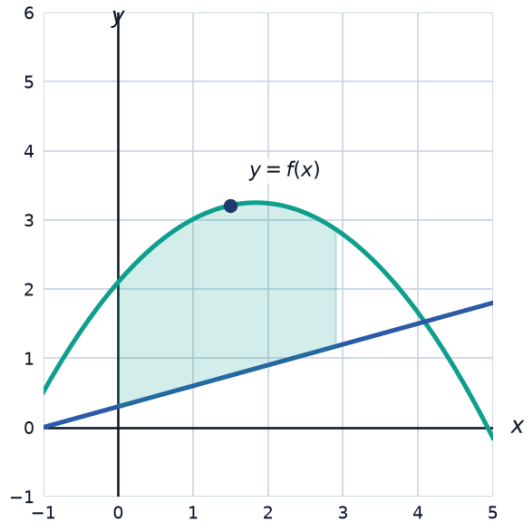
b. $\frac{177\pi}{2}$

c. $\frac{179\pi}{2}$

d. $\frac{225\pi}{2}$

23 [Foundational] • Target: Area between two polar curves

Find the area of one petal of $r = 4\cos(2\theta)$.



a. $\frac{9\pi}{4}$

b. 2π

c. 4π

d. 8π

24 [Foundational] • Target: Area of one petal of a rose curve

Find the area inside the circle $r = 9$ but outside the circle $r = 2$, for $\theta \in [0, \pi]$ only (half the full region).

a. 77π

b. $\frac{79\pi}{2}$

c. $\frac{77\pi}{2}$

d. $\frac{49\pi}{2}$

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25 [Foundational] • Target: Setting up area between curves that intersect

Consider the parametric curve defined by $x(t) = 3\cos(t) - \cos(3t)$ and $y(t) = 3\sin(t) - \sin(3t)$ for $t \in [0, 2\pi]$. This curve traces out a shape known as a 'cardioid'. Find the maximum area of a triangle that can be inscribed within the loop of this cardioid, with one vertex at the origin.

a. $\frac{27\sqrt{3}}{2}$

b. $\frac{81}{2}$

c. $\frac{81\sqrt{3}}{4}$

d. $54\sqrt{3}$

26 [Foundational] • Target: Area of a full rose curve

Polar area between two curves $r_1(\theta) \le r_2(\theta)$:

a. $\frac{1}{2}\int (r_2^2 - r_1^2)d\theta$

b. $\frac{1}{2}\int (r_2 - r_1)^2 d\theta$

c. $\int (r_2 - r_1)d\theta$

d. $\pi(r_2^2 - r_1^2)$

27 [Foundational] • Target: Area inside a limaçon

Consider a region in the first quadrant bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis. A rectangle is inscribed within this region such that one vertex is at the origin, two sides lie along the x and y axes, and the fourth vertex lies on the curve $y = 9 - x^2$. If the area of this inscribed rectangle is maximized, determine the value of the definite integral that represents the volume of a solid formed by revolving the region bounded by the curve $y = 9 - x^2$, the x-axis, and the y-axis about the y-axis.

a. $\frac{81\pi}{2}$

b. 81π

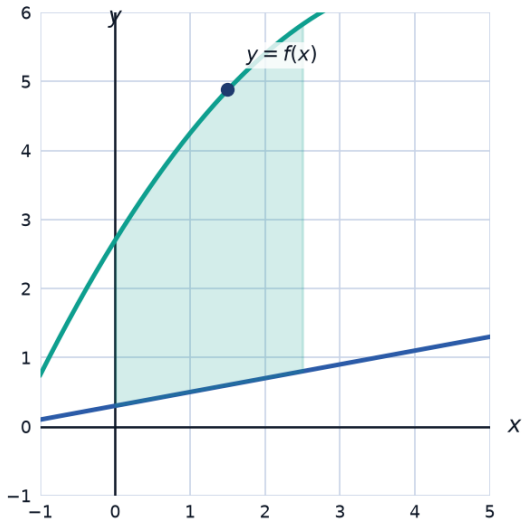
c. 162π

d. $\frac{243\pi}{2}$

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28 [Foundational] • Target: Setting up area problems (AP-style, no evaluation)

A rectangular garden is to be enclosed by a fence. One side of the garden is adjacent to a long straight river, and requires no fencing. The other three sides require fencing, and the total length of the fence available is 120 meters. Let the side of the garden parallel to the river have length x meters, and let the two sides perpendicular to the river each have length y meters. The area of the garden is given by $A(x, y) = xy$. \\ Find the dimensions x and y that maximize the area of the garden.



a. x = 60 m, y = 30 m

b. x = 30 m, y = 60 m

c. x = 40 m, y = 40 m

d. x = 80 m, y = 20 m

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding intersection points of polar curves

Find the area of one petal of the rose $r = 1\cos(3\theta)$ (3 odd).

a. $\frac{\pi}{12}$

b. $\frac{\pi}{6}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{24}$

30 [Intermediate] • Target: Area using symmetry to simplify computation

Area of one petal of $r = 2\sin 3\theta$:

a. $\pi/3$

b. $2\pi/3$

c. π

d. $\pi/6$

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31 [Intermediate] • Target: Area review: full worked example

Limaçon $r = a + b\sin\theta$. For $a = b = 1$ (cardioid), area:

a. $\pi/2$

b. $3\pi/2$

c. 2π

d. π

32 [Intermediate] • Target: Comparing polar area to rectangular area (conceptual)

Polar area $A = \frac{1}{2}\int r^2\,d\theta$. For one petal of $r = \sin 2\theta$:

a. $\pi/4$

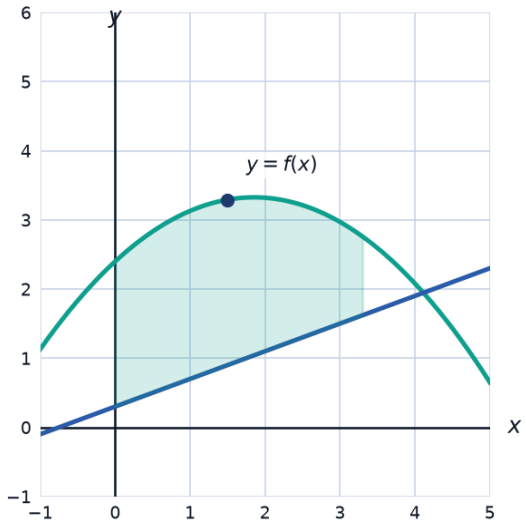
b. $\pi/8$

c. $\pi/2$

d. 1

33 [Intermediate] • Target: The polar area formula

The area of the region enclosed by $x = t - \sin t, y = 1 - \cos t$ (one arch of cycloid) using $A = \int y\,dx$:



a. πr^2

b. $3\pi r^2$ (with $r = 1$)

c. π

d. 2π

34 [Intermediate] • Target: Area of a full cardioid

Area swept inside one loop of $r^2 = 4\sin 2\theta$:

a. 1

b. 2

c. 4

d. π

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35 **[Intermediate]** • **Target: Area between two polar curves**

Area of region bounded by parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (full astroid):

a. $3\pi/8$

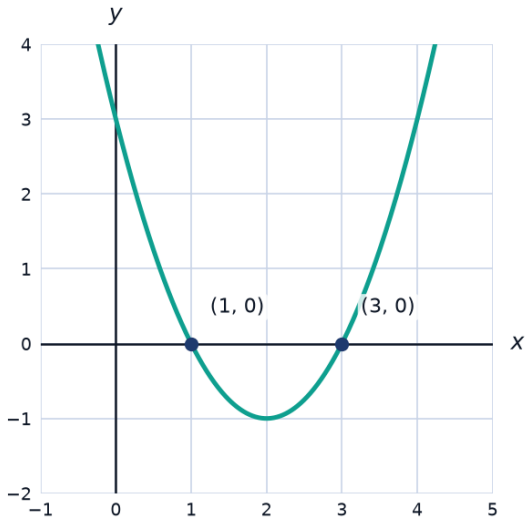
b. $3\pi/4$

c. $\pi/4$

d. $\pi/2$

36 **[Intermediate]** • **Target: Area of one petal of a rose curve**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points where $f(x) = 0$. Determine the sum of the x-coordinates of these intersection points.



a. 1

b. 3

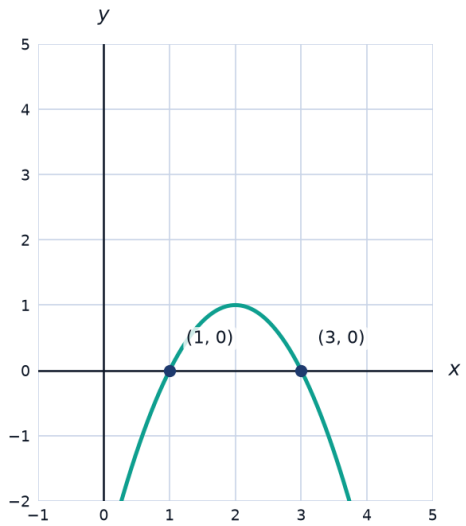
c. 4

d. -3

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37 [Intermediate] • Target: Setting up area between curves that intersect

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?



a. 1

b. 2

c. 3

d. 4

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Area of a full rose curve • Cross-Lesson: 3.01 Parametric Equations

If $|f^{(4+1)}(c)| \le 1$ on the relevant interval, bound the Lagrange error $|R_4(x)|$ for approximating f at $x = 0.1$ using a Taylor polynomial centred at $a = 1$.

a. 0.0098415

b. 0.00492075

c. 0.00246038

d. 0.0273375

39 [Challenging] • Target: Area inside a limaçon • Cross-Lesson: 3.02 Vector-Valued Functions

Suppose $|f^{(3+1)}(c)| \le 3$ for all c between 1 and 0.1. Bound $|R_3(0.1)|$ for the Taylor polynomial centred at $a = 1$. (extra variant)

a. 0.0410063

b. 0.0820125

c. 0.3645

d. 0.164025

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40 [Challenging] • Target: Setting up area problems (AP-style, no evaluation) • Cross-Lesson: 4.01 Polar Curves and Derivatives

Euler's method approximates solutions by using:

a. Second derivatives

b. Tangent line steps

c. Exact antiderivatives

d. Taylor polynomials of degree 2

41 [Challenging] • Target: Finding intersection points of polar curves • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

For $\frac{dy}{dx} = 2y,\ y(0) = 1$, use one Euler step of size $h = 0.25$ to approximate $y(0.25)$.

a. 1.5

b. 0.5

c. 2

d. 1.25

42 [Challenging] • Target: Area using symmetry to simplify computation • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For $\frac{dy}{dx} = x + y,\ y(0) = 1$, take one Euler step of size $h = 0.5$ to approximate $y(0.5)$.

a. 1

b. 1.5

c. 2

d. 1.55

43 [Challenging] • Target: Area review: full worked example • Cross-Lesson: 6.01 Power Series and Interval of Convergence

A rectangular field is to be enclosed by a fence. One side of the field is adjacent to a river and does not require fencing. The farmer has 120 meters of fencing material. Let the dimensions of the field be x meters perpendicular to the river and y meters parallel to the river. The area of the field is given by $A = xy$. Part A: Express the area A as a function of x. Part B: Determine the dimensions x and y that maximize the area of the field. Provide the maximum possible area.

a. $x = 20$ m, $y = 80$ m, $A_{\max} = 1600$ m^2

b. $x = 30$ m, $y = 60$ m, $A_{\max} = 1800$ m^2

c. $x = 40$ m, $y = 40$ m, $A_{\max} = 1600$ m^2

d. $x = 25$ m, $y = 70$ m, $A_{\max} = 1750$ m^2

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44 [Challenging] • Target: Comparing polar area to rectangular area (conceptual) • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Find the area inside $r = 2\cos(\theta)$ and outside $r = 1$.

a. π

b. $\sqrt{3} + \frac{2\pi}{3}$

c. $\frac{\sqrt{3}}{2} + \frac{\pi}{3}$

d. $\frac{\pi}{3}$

45 [Challenging] • Target: The polar area formula • Cross-Lesson: 1.01 Integration by Parts

Which formula gives the length of a polar curve $r = f(\theta)$?

a. $\int_{\alpha}^{\beta} r\,d\theta$

b. $\int_{\alpha}^{\beta} |dr/d\theta|\,d\theta$

c. $\int_{\alpha}^{\beta} r^2\,d\theta$

d. $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$

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